



CricMate

From ball to broadcast, instantly

AMEY PANDIT (AIML)

ANIKET PARATE (AIML)

ASHUTOSH PATIL (JS)

BABLU GUPTA (DB)

WHAT ARE WE SOLVING FOR?

We aim to revolutionize cricket content delivery with AI-driven automation.



Problem

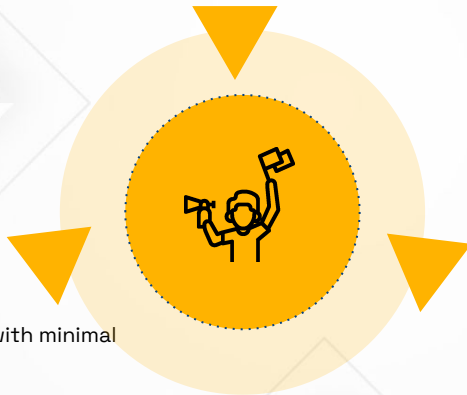
Cricket coverage today suffers from a fragmented, manual pipeline. The current cricket data pipeline runs from manual scoring at the ground to tagging, processing, and editing before producing highlights, reels, and insights. Each stage is human-dependent, slow, and costly, greatly affecting the rate of production and delivery.



Solution

Automated Workflows – Scoring, commentary transcription and generation, and highlights with minimal manual effort.

Allow the people who act as the bridge between the broadcaster and fan to have more granular access to creating faster content – Creating a platform that allows for automation of the essential workflows that are involved in delivery to the end user.



Key Impact / KPIs

60-70%+ reduction in manual scoring effort – automated event detection replaces traditional scorers.

On the go quick content creation– Highlight reels and packages, images as well as narration can be created and done instantaneously..

2x-3x operational efficiency – same team can manage multiple matches simultaneously.

Improvement in turn around time – automated workflows will lead to a rapid improvement in TAT.

- **Manual scoring dependency** – Cricket data collection during a live match remains human-dependent, making the process slow and resource-heavy.
- **Slow processing pipeline** – Tagging, editing, and highlight generation involve multiple manual steps, delaying timely delivery.
- **High operational costs** – Human-driven workflows require large teams, leading to high production and broadcasting expenses.
- **Inefficient content creation & delivery** – Highlights, reels, and insights are produced manually, limiting instant content generation and real-time fan engagement.

Automated Scoring & Commentary – Use AI-powered workflows to capture live scoring where we collect batting, bowling and fielding details of a ball in a live match automatically, transcribe commentary, and even generate real-time insights automatically, reducing dependence on manual input.

Instant Highlight Generation – Enable automated clipping and editing of key match moments (boundaries, wickets, milestones) into a concise action packed fast package.

Empowering Content Teams – Provide broadcasters, analysts, and media teams with tools that allow granular control over automated workflows, so they can quickly customize and deliver content.

Unified Delivery Platform – Create a centralized platform that integrates scoring, commentary, highlights, and analytics, ensuring faster, seamless content flow to TV, OTT, and social media platforms.

Enhanced Fan Engagement – Deliver real-time, high-quality content directly to fans across multiple channels, boosting engagement, reach, and overall viewer experience in their preferred language.

KEY IMPACT/ KPIS



60-70%+ reduction in manual scoring effort – Automated event detection captures runs, wickets, and match progress in real-time, replacing traditional scorers and freeing human effort for higher-value tasks. Every piece of content that is generated via a cricket match involves human dependency which in turn creates a lot of other issues that one might encounter in a manual workflow due to human error like getting delayed streams, incorrect stream switching, etc.

Instant, on-the-go content creation – Highlight reels, image packages, and even auto-narrated clips can be generated within seconds, ensuring fans receive engaging content while the action is still unfolding.

2x–3x operational efficiency – A single production team can seamlessly manage multiple matches simultaneously, lowering costs while dramatically scaling coverage.

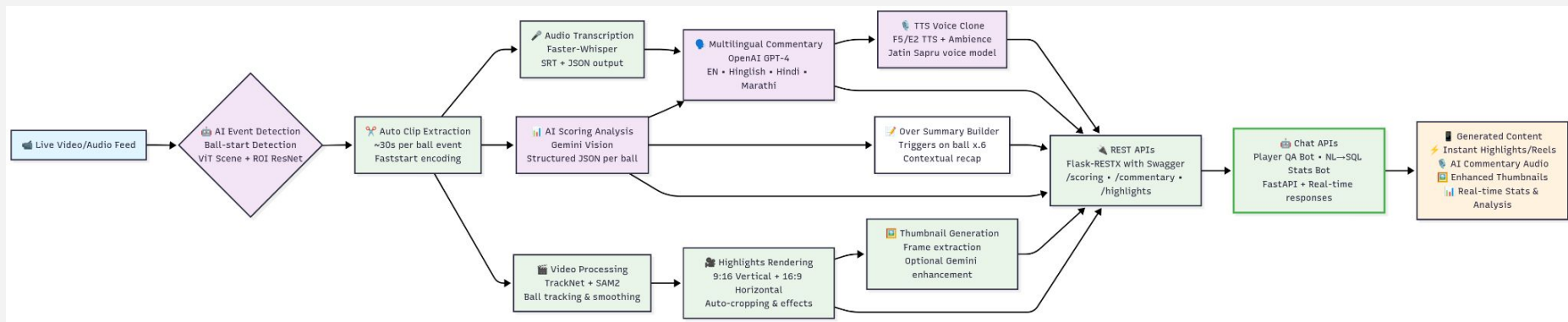
Rapid turnaround time (TAT) – Automated workflows eliminate bottlenecks in editing and packaging, cutting hours of post-production into minutes for near real-time delivery.

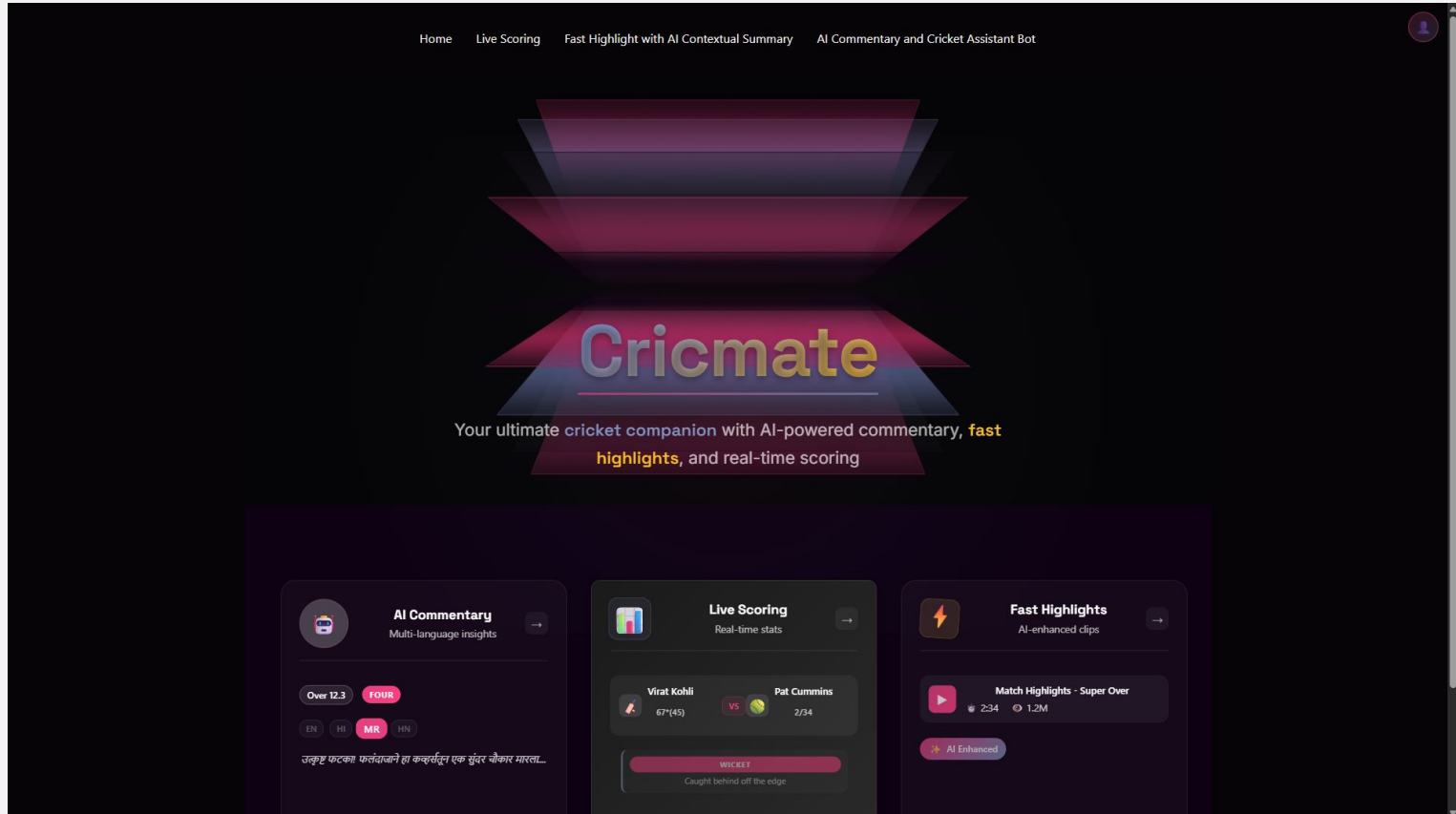
Improved Efficiency and Cost Savings – Streamlined workflows and automation enable faster operations while reducing overall production expenses.

Easy to Scale – Automated systems can be effortlessly replicated across multiple matches, leagues, or geographies without needing proportional increases in manpower or resources. This flexibility allows broadcasters and sports organizations to expand coverage rapidly while maintaining consistent quality and efficiency.

Faster Multi-Sport Adaptation – The same automation framework can be quickly tailored to different sports, enabling rapid deployment across cricket, football, basketball, or any other game, ensuring broader reach and maximized ROI with minimal reconfiguration.

FLOW DIAGRAM





[Home](#)[Live Scoring](#)[Fast Highlight with AI Contextual Summary](#)[AI Commentary and Cricket Assistant Bot](#)

Live Scoring

Over 0.1

DOT BALL

View →



BATTER: Bethell

NON-STRIKER: Kohli

BOWLER: Khaleel

BOWLING STYLE: Left-arm Fast-medium

Over 0.2

TWO RUNS

View →



BATTER: Bethell

NON-STRIKER: Kohli

BOWLER: Khaleel

BOWLING STYLE: Left-arm Fast-medium

Over 0.3

FOUR

View →



BATTER: Bethell

NON-STRIKER: Kohli

BOWLER: Khaleel

BOWLING STYLE: Left-arm fast-medium

Over 0.4

FOUR

View →



BATTER: Bethell

NON-STRIKER: Kohli

BOWLER: Khaleel

BOWLING STYLE: Left-arm Fast-medium

Over 0.5

FOUR

View →



BATTER: Bethell

NON-STRIKER: Kohli

BOWLER: Khaleel

BOWLING STYLE: Left-arm Fast-medium

Over 0.6

SINGLE

View →



BATTER: Bethell

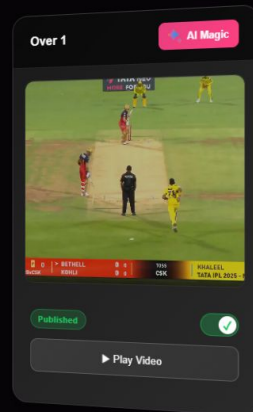
NON-STRIKER: Kohli

BOWLER: Khaleel

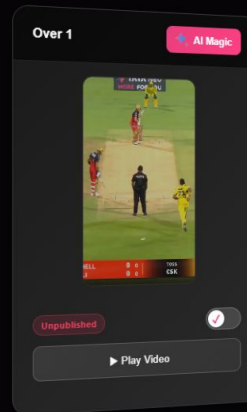
BOWLING STYLE: Left-arm fast-medium

Fast Highlight with AI Contextual Summary

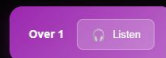
Horizontal Videos (16:9)



Vertical Videos (9:16)



 AI Audio Commentary



[Home](#) [Live Scoring](#) [Fast Highlight with AI Contextual Summary](#) [AI Commentary and Cricket Assistant Bot](#)



Fast Highlight with AI Contextual Summary

Horizontal Videos (16:9)

Over 1

AI Magic

AI Generated



Published

☒

▶ Play Video

Vertical Videos (9:16)

Over 1

AI Magic



Unpublished

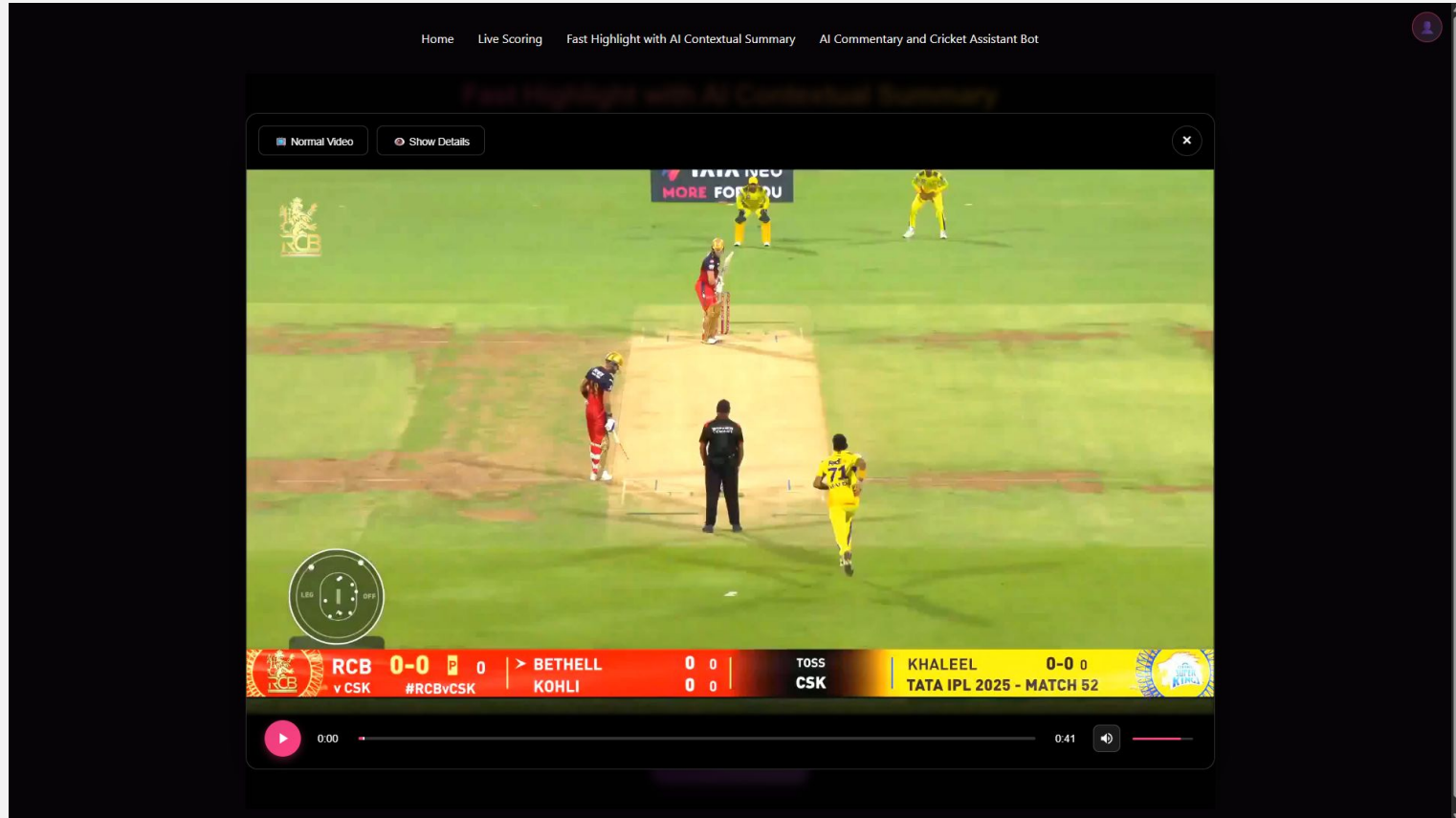
☐

▶ Play Video

AI Audio Commentary

Over 1

 Listen



[Home](#) [Live Scoring](#) [Fast Highlight with AI Contextual Summary](#) [AI Commentary and Cricket Assistant Bot](#)

AI Commentary and Cricket Assistant Bot



Over
0.1

DOT
BALL

Unpublished



Khaleel ne achhi gend daali, Bethell ne off side pe carefully defend kiya. Pehla over, pehli gend, abhi tak koi run nahi bana, RCB abhi bhi 0 pe hai.

EN

HI

MR

HN



Over
0.2

TWO
RUNS

Published



Bethell ne ekdam swift flick shot khela, aur mid-wicket mein gap dhoond liya! Ye toh do run ho gaye! Power play ke leading wicket-taker Khaleel bhi nahi rok paye isse.

EN

HI

MR

HN



Over
0.3

FOUR

Published



Khaleel ki delivery short aur wide, Jacob Bethell ne mauka dekha aur cut shot ke saath ball ko point boundary par bhej diya. Kya mazaakia shot tha woh!

EN

HI

MR

HN



Over
0.4

FOUR

Unpublished



Bethell ne late cut kiya, aur ball third man ki taraf race kar rahi hai. Khaleel, jo left-arm fast-medium bowler hai, chaunk gaye honge jab ball fielder ke haath mein kam se kam 100 ya 150 ya 200 pe hai.

EN

HI

MR

HN



Home Live Scoring Fast Highlight with AI Contextual Summary AI Commentary and Cricket Assistant Bot

AI Commentary and Cricket Assistant Bot



Over 0.1

DOT BALL

Unpublished

✓

EN

HI

MR

HN

खलील, बाएं हाथ के तेज-मध्यम गेंदबाज, ने शानदार डिलीवरी के साथ इनिंग्स की शुरुआत की, बेथेल ने इसे ऑफ साइड पर सतर्कता से डिफेंड किया। पहले ओवर की पहली गेंद, अब तक कोई रन नहीं बना, RCB अभी भी 0 पर है।



Over 0.2

TWO RUNS

Published

✓

EN

HI

MR

HN

दाएं हाथ के बेदसमन बेथेल ने तेजी से एक फ्लिक शॉट खेला! उन्होंने मिड-विकेट में एक गैप खोज लिया, और ये दो रन स्कोरबोर्ड में जोड़ दिए! खलील, पॉवर प्ले में सबसे अधिक विकेट लेने वाले, इसे रोक नहीं सके।



Over 0.3

FOUR

Published

✓

EN

HI

MR

HN

खलील ने एक छोटी गेंद डाली, लेकिन यह बहुत चौड़ी थी। बाएं हाथ के बल्लेबाज जेकब बेथेल ने इस पर झूम कर हमला बोला और खतिशाली कट शॉट के साथ इसे प्वाइंट क्षेत्र की ओर दौड़ते हुए भेज दिया। यह एक चौका था!



Over 0.4

FOUR

Unpublished

✓

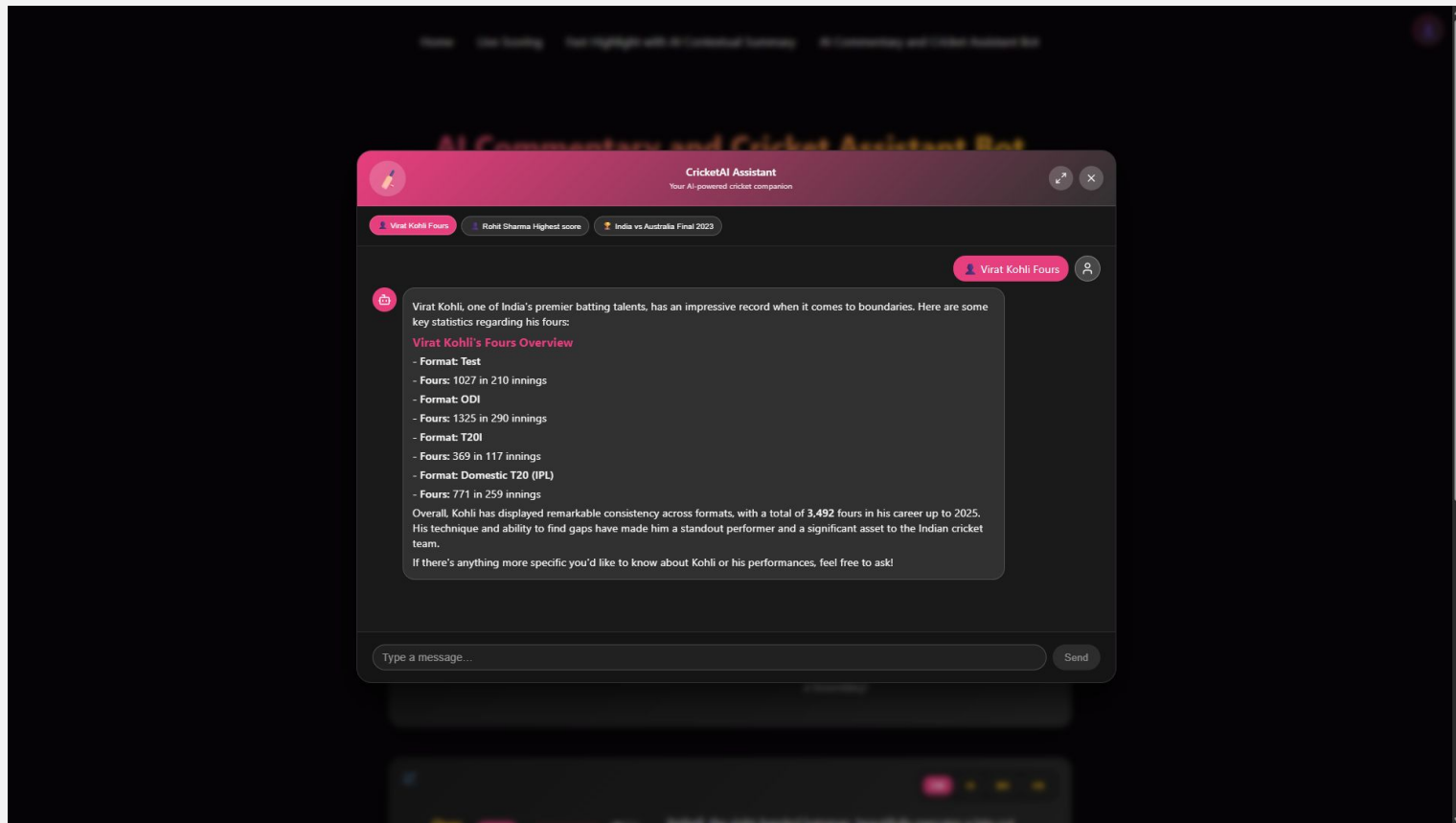
EN

HI

MR

HN

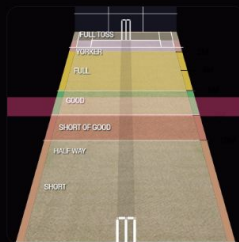
बेथेल ने लेट कट की खूबसूरती से निभाई, और गेंद तीसरे आदमी की ओर दौड़ने लगी। खलील, जो बाएं हाथ के तेज-मध्यम गेंदबाज हैं, चौक गए होंगे जब गेंद प्वाइंट क्षेत्र की ओर दौड़ती है। RCB अभी 4 रन पर है।



[Home](#)
[Live Scoring](#)
[Fast Highlight with AI Contextual Summary](#)
[AI Commentary and Cricket Assistant Bot](#)

[← Back](#)

Over 0.1 – RCB 0/0



Batting

BETHELL

On Strike

0 RUNS 1 BALLS 0 4S 0 SR
 Right Handed Bat

KOHLI

Not Out

0 RUNS 0 BALLS 0 4S 0 SR

Bowling

KHALEEL

0.1 overs • 0 runs • 0 wickets
 Left-arm Fast-medium
 Yorker

Shot Details

Defensive

Push • Bat • Forward

Off Side

Pitch: GOOD

0 runs

Commentary

DOT BALL

Just confirming for all you RCB fans who are a little wondering why the mighty Josh Hazlewood isn't playing today, we're hearing it's a shoulder niggle. So first game for Lungi Ngidi.

Fielding

No specific fielding positions noted

MVP PLAN

Week 1: Foundation & Research

Days 1-2: Research & Analysis

- **Market Research:** Map manual scoring workflows, broadcaster bottlenecks, and human error challenges (e.g., delayed updates, incorrect event logging, slow highlight generation).
- **Dataset Collection:** Curate cricket video samples for model training/testing (ODI, T20, Test matches).
- **Technical Architecture:** Finalize modular design ensuring scalability to multi-match coverage.
- **Baseline Metrics:** Document manual TAT for scoring, highlights, and commentary.

Days 3-5: Core AI Models Setup

- **Ball Detection Model:** Fine-tune ViT + ResNet for run/wicket/event detection.
- **TrackNet Integration:** Set up ball trajectory & impact tracking pipeline.
- **Gemini API:** Configure structured JSON outputs for automated scoring.
- **Whisper Setup:** Deploy multilingual transcription pipeline for on-field chatter & umpire audio.

Days 6-7: Infrastructure

- **Environment:** Docker containers, GPU orchestration, CI/CD pipelines.
- **Database Schema:** SQLite → scalable to Postgres for match sessions, players, events, artifacts.
- **API Framework:** Flask-RESTX skeleton with Swagger for early endpoint testing.

Week 2: Core Pipeline Development

Days 8-10: Video Processing Pipeline

- **Auto-Clipping Service:** 30s ball-wise clip extraction.
- **Scene Classification:** ViT-powered cricket context recognition.
- **SAM2 Integration:** Ball tracking refinement with segment-anything models.
- **Video Rendering:** Auto-generate 9:16 (mobile-first) + 16:9 (broadcast) highlights.

Days 11-12: AI Content Generation

- **Multilingual Commentary:** GPT-4 powered (English, Hindi, Tamil, Bengali).
- **Over Summaries:** Auto-trigger summary generation on every ball x.6.
- **TTS Voice Cloning:** Human-like voice cloning with F5/E2 for authentic commentary feel.
- **Thumbnail Generator:** Frame extraction + enhancement for reels/stories.

Days 13-14: API Development

- **REST Endpoints:** /scoring, /commentary, /highlights, /streaming.
 - **Chat APIs:** Player QA bot + NL→SQL stats bot for instant insights.
 - **Error Handling:** Retry logic, fallback responses, and failover pipelines.
- Performance:** Async video processing + queuing for scale.

Week 3: Integration & Testing

Days 15-17: End-to-End Integration

- **Pipeline Orchestration:** Connect ball detection → scoring → highlights → commentary.
- **Real-time Processing:** Ingest live video streams with low-latency outputs.
- **QA:** Unit + integration testing across all modules.
- **Benchmarking:** Compare AI vs manual TAT (time per ball/over).

Days 18-19: MVP Validation

- **Efficiency Validation:** Achieve **60-70%+ manual effort reduction**.
- **Content Quality Review:** Evaluate multilingual commentary + highlight reels.
- **API Testing:** Load/stress testing with simulated multi-match coverage.
- **User Testing:** Internal stakeholders simulate real-world workflows.

Days 20-21: Deployment & Documentation

- **Production Setup:** AWS/GCP/K8s deployment.
- **API Docs:** Swagger + developer handbooks.
- **User Guides:** Scorer/operator quick-start playbook.
- **Demo Prep:** Showcase full pipeline on real cricket match footage.

Top risks & mitigations



1. **AI Accuracy Issues** → Wrong scoring or missed events; solved with ensembles, retraining & human validation.
2. **Real-time Delays** → Latency in highlights/live scoring; mitigated by GPU acceleration & async scaling.
3. **API Rate Limits** → External service caps; fallback to local models, batching & multi-provider setup.
4. **Data Quality & Scalability** → Poor feeds or infra overload; use multi-source inputs & distributed microservices.
5. **User Adoption & Content Quality** → Resistance to AI or “non-authentic” outputs; managed via training, ROI proof & A/B testing.
6. **Compliance (Privacy & IP)** → Risks of data leaks or licensing issues; ensure encryption, access controls & legal reviews.

Ask (Support Needed)



1. **Approvals & Access** → Pilot partner / rights-holder match footage for training + MVP validation.
2. **AIML Training Pipelines** → Support for dataset curation, labeling resources, GPU infra for model training.
3. **Resources** → 1 infra mentor, ~1 hr/week engineering review support, small pilot budget for infra/cloud costs.
4. **Integration Support** → Guidance on broadcaster CMS integration, social media publishing APIs, and real-time feeds.
5. **Monitoring & MLOps** → Tools for performance monitoring, model drift detection, and experiment tracking.
6. **Goal** → Enable robust MVP delivery with real-world validation, scalable infra, and faster adoption.



APPENDIX

[DOC LINK](#)



THANK YOU